Econ 527 HW1

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April 2026

1 Neoclassiccal Growth Model

We seek to solve a simple represenative agent neoclassical growth model. An agent solves

$$\begin{aligned} \max_{\{c_t\}_{t=0}^{\infty}} \ & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & k_{t+1} = Ak_t^{\alpha} + (1-\delta)k_t - c_t \end{aligned} \tag{1}$$

where $u(c) = \log(c)$, $k_t > 0$, and k_0 is given. The problem 1 can be written as

$$V(k) = \max_{k' \geq 0} \log\left(Ak^{\alpha} + (1-\delta)k - k'\right) + \beta V\left(k'\right) \tag{2}$$

We set the following parameters: $\beta = 0.96, A = 1, \delta = 1$, and $\alpha = 0.35$.

1. Analytically, we make initial guess

$$V_0 = a + b \log k \tag{3}$$

, (2) transforms into

$$a + b\log k = \max_{k' \geq 0} \log(Ak^{\alpha} + (1-\delta)k - k') + \beta[a + b\log k']$$

and taking the first-order condition on k'

$$\begin{aligned} 0 &= \frac{-1}{Ak^{\alpha} + (1-\delta)k - k'} + \beta\frac{b}{k'} \\ \beta b\left[Ak^{\alpha} + (1-\delta)k - k'\right] &= k' \\ \implies k'(k) &= \frac{\beta b[Ak^{\alpha} + (1-\delta)k]}{1+\beta b} \end{aligned} \tag{4}$$

Substituting the policy function 4 into the Equation 2 gets

$$\begin{aligned} V(k) = \log\Bigg(Ak^{\alpha} + (1-\delta)k - \left[\frac{\beta b[Ak^{\alpha} + (1-\delta)k]}{1+\beta b}\right]\Bigg) \\ + \beta V\left(\left[\frac{\beta b[Ak^{\alpha} + (1-\delta)k]}{1+\beta b}\right]\right) \end{aligned}$$

Since $\delta = 1$, this simplifies to

$$\begin{aligned} V(k) &= \log\left(Ak^\alpha - \left[\frac{\beta b[Ak^\alpha]}{1+\beta b}\right]\right) + \beta V\left(\frac{\beta b[Ak^\alpha]}{1+\beta b}\right) \\ &= \log\left(\frac{Ak^\alpha}{1+\beta b}\right) + \beta V\left(\frac{\beta b[Ak^\alpha]}{1+\beta b}\right) \end{aligned}$$

substituting V_0

$$\begin{aligned} a + b\log k &= \log\left(\frac{Ak^\alpha}{1+\beta b}\right) + \beta\left(a + b\log\left[\frac{\beta b[Ak^\alpha]}{1+\beta b}\right]\right) \\ &= \log\left[\left(\frac{Ak^\alpha}{1+\beta b}\right)\left(\frac{\beta b[Ak^\alpha]}{1+\beta b}\right)^{\beta b}\right] + \beta a \end{aligned}$$

guessing coefficients has

$$\begin{aligned} b &= \alpha + \alpha\beta b \\ \implies b &= \frac{\alpha}{1-\alpha\beta} \end{aligned}$$

since $\log Ak^\alpha = \log A + \alpha \log k$ and

$$\begin{aligned} a &= \log\left[\left(\frac{A}{1+\beta b}\right)\left(\frac{A\beta b}{1+\beta b}\right)^{\beta b}\right] + \beta a \\ a &= \frac{1}{1-\beta}\log\left[\left(\frac{A}{1+\beta b}\right)\left(\frac{A\beta b}{1+\beta b}\right)^{\beta b}\right] \end{aligned}$$

and evaluated with the current parameters gives

$$a \approx -24.03 \quad b \approx 0.53$$

2. For this section the algorithm is

Algorithm 1 Value Function Iteration with Spline Interpolation

Require: Grid $\{k_i\}_{i=1}^N$, parameters β, α, δ, A, tolerance ϵ

```
Initialize V^0(k_i) = 0 for all i, set n = 0
repeat
    for each grid point k_i do
        Construct cubic spline interpolant Ṽ^n from {k_i, V^n(k_i)}
        Solve V^{n+1}(k_i) = max_{k'≥0} { log(A k_i^α − k') + β Ṽ^n(k') }
        Record optimizer k'*(k_i)
    end for
    err = max_i |V^{n+1}(k_i) − V^n(k_i)|
    n ← n + 1
until err < ε(1 − β)
```

Ensure: Value function V^n and policy function k'^*

Running the numerical solution gives

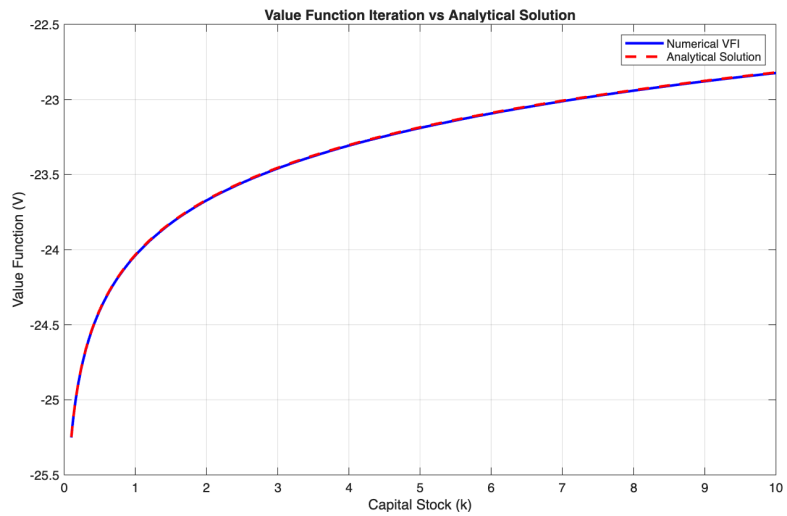


Figure 1: Analytical vs Numerical Solution to VFI

The maximum error between the analytic and numerical compute policy functions are 0.027.

3. The top of the wealth grid does not bind in this problem for any state (i.e. there is no k' such that $k^{*\prime} \geq \overline{k} = 10$, and the agent starts de-accumulating wealth at $k = 0.02$.

4. From this analysis, it took 306 iterations taking about 1.7 seconds.

2 Income Fluctuation Problem

1. We seek to solve

$$V(a,z) = \max_{a' \in [a_{\min}, a_{\max}]} \frac{\left[(1+r)a + \exp(z) - a'\right]^{1-\gamma}}{1-\gamma} + \beta \mathrm{E}\left[V\left(a', z'\right) \mid z\right]$$
$$z' = \rho z + \varepsilon, \varepsilon \sim N\left(0, \sigma^2\right)$$

using the following

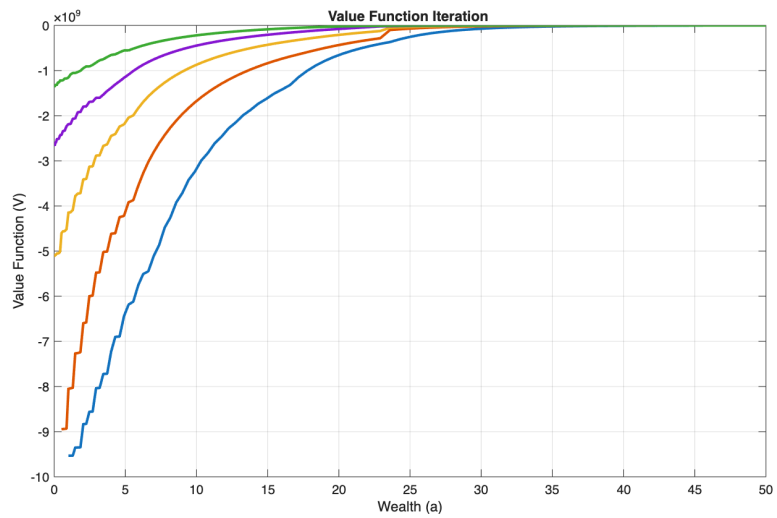
Algorithm 2 Income Fluctuation Value Function Iteration with Spline Interpolation

Require: Grids $\{a_i\}_{i=1}^{N_a}$ and $\{z_j\}_{j=1}^{N_z}$, parameters $r, \gamma, \beta, \rho, \sigma$, tolerance ϵ, and transition matrix Π

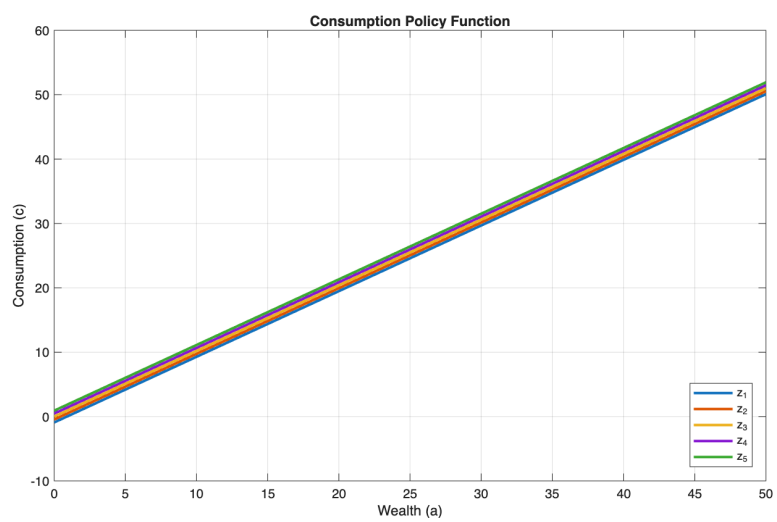
```
Initialize V^0(a_i, z_j) = 0 for all i, j, set n = 0
repeat
    for each grid point (a_i, z_j) do
        Solve V^{n+1}(a_i, z_j) = u(c) + β E[Ṽ^n(z', a')|z] using Π
        for each income z_j do
            Construct cubic spline interpolant Ṽ^n from {a_i, z_j, V^n(a_i, z_j)}
        end for
        Record optimizer k'
    end for
    err = max_{i,j} |V^{n+1}(a_i, z_j) − V^n(a_i, z_j)|
    n ← n + 1
until err < ε(1 − β)
```

Ensure: Value function V^n and policy function k'^*

2. The value function (Figure 2a) and policy function (Figure 2b) can be seen below.



(a) Value Function



(b) Policy Function

Figure 2: VFI results for Income Fluctuation Problem

Carroll 1997 says that consumer saving behavior can be better characterized by a "buffer-stock" model than the traditional Permanent Income Hypothesis by setting average consumption growth equal to average income growth i.e. $\Delta \ln E_{,t} C_{i,t+1} = \Delta \ln E_{,t} P_{i,t+1}$, leading to unique and stable ratio between cash on hand (wealth plus disposable income) and the

permanent component of income (Jappelli, Padula, and Pistaferri 2008). These results also imply the existence of a target wealth stock, stemming from the competing impatience and prudent behaviors of consumers.

Since the 45 degree line would be a little difficult to see in Figure 2b, we plot $a' - a$ in Figure 3. If wealth is below target, then consumers will save and if wealth is above target, consumers will draw down their wealth stocks.

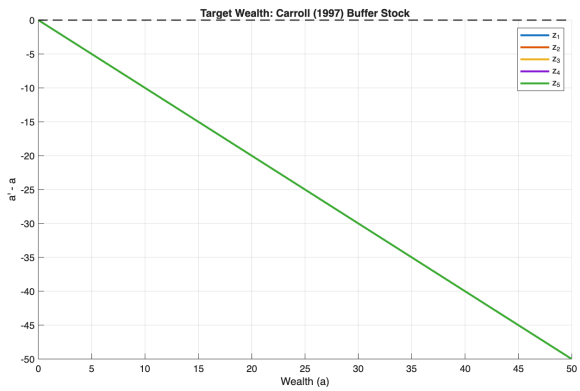


Figure 3: Target Wealth on Welath Grid

All values of a are above target, so to achieve $a' - a = 0$, consumers must dis-save i.e. consume. This matches the policy function recommendation. Thus, in the face of uncertainty and consumers in this model draw down their wealth until $a' - a = 0$.

3. The wealth grid is binding at the lower bound. This problem took 568 iterations and took around 62 seconds to solve.

3 Description of Code

- `PS2_Q1.m` - code for solving the Necolassical Growth Model (Q1)
- `PS2_Q2.m` - code for solving the Income Fluctuation Problem (Q2)
- `PS2_Q2_jax.ipynb` - attempted to redo Q2 in python to check my work but they don't have built in spline and I fixed my Matlab code anyways
- `polynomial_grid.m` - a file with function to make polynomial grids
- `rouwenhorst.m` - script to discretize using Rouwenhorst's method (courtesy of Professor Greaney)

- `cubic_spline_interpolation.m` - function to create cubic spline interpolation
- `PS2.pdf` - the assignment sheet

References

Carroll, Christopher D. (Feb. 1997). *Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis on JSTOR*. URL: https://www.jstor.org/stable/2951275 (visited on 04/21/2026).

Jappelli, Tullio, Mario Padula, and Luigi Pistaferri (Dec. 2008). "A Direct Test of The Buffer-Stock Model of Saving". In: *Journal of the European Economic Association* 6.6, pp. 1186–1210. ISSN: 1542-4766, 1542-4774. DOI: 10.1162/JEEA.2008.6.6.1186. URL: https://academic.oup.com/jeea/article-lookup/doi/10.1162/JEEA.2008.6.6.1186 (visited on 04/29/2026).